Dual Learning for Cross-Domain Recommendations: Improving Efficiency and Effectiveness of Recommender Systems

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Abstract

Cross-domain recommender systems have been widely adopted by many companies to enhance user experiences and have been crucial to success in many business applications. However, previously proposed cross-domain models only aim at utilizing information from the source domain to improve recommendations in the target domain, while not considering bidirectional nature of the cross-domain recommendation task. Note, however, that the target domain can also be used to improve recommendations in the source domain. In addition, existing models often require a large set of overlap users for the learning process, which is difficult and expensive to obtain in practice.

To address these concerns, in this paper we propose to provide cross-domain recommendations based on the mechanism of dual learning that transfers information between two related domains in an iterative manner until the learning process stabilizes. Specifically, we developed a novel latent orthogonal mapping to extract user preferences over multiple domains while preserving relations between users across different latent spaces. By combining dual learning with the autoencoder method that extracts the latent essence of feature information, the proposed approach is capable of improving efficiency and effectiveness of cross-domain recommendations significantly, as we show in the paper via extensive experiments on real-world datasets.

Introduction

Recommender systems have become a crucial component for online marketplaces, and their success is due to a large extent to the collaborative filtering (CF) technologies (Sarwar et al. 2001) that help users to discover relevant products. Among the key challenges of the CF systems are the data sparsity and the cold start problems (Adomavicius & Tuzhilin 2005) because it is hard to understand and model user preferences accurately when we have access only to a small fraction of user transactions.

To address these problems, researchers propose to use cross-domain recommendation methods (Cantador et al. 2015; Hu et al. 2018; Lian et al. 2017; Sahebi et al. 2017) through transfer learning (Pan & Yang 2009) that learn user preferences in the source domain and transfer them to the target domain. For instance, if a user watches a certain movie, we can recommend the original novel on which the movie is based to that user to read. However, the previously proposed cross-domain recommendation methods have only focused on the unidirectional transfer learning from the source domain to the target domain, while not considering the bidirectional nature of cross-domain recommendations. This is unfortunate because the target domain could also be used to improve recommendations in the source domain in a mutually beneficial manner. For example, once we know the type of books that the user would like to read, we can recommend movies on related topics and vice versa, thus forming a bi-directional loop across the two domains in order to provide better recommendations in both domains.

This bi-directional approach is known as dual learning (Long et al. 2012), and it has been studied in multiple learning settings and applied to different domains, including machine translation, sentimental analysis, image classification and question answering systems (He et al. 2016). However, it has not been used in recommender systems so far,

and we propose to apply dual learning for the cross-domain recommendation purposes.

Note, however, that to implement successful dual learning recommender systems, it is crucial to have a significant overlap of user information across the two domains for providing satisfying cross-domain recommendations, where these users have consumed items in both categories, such as watching a movie and reading a book. These `co-purchase' behaviors for the overlap users are used as `pivots' to learn the relations of user preferences in different domains. However, the requirement of large numbers of overlap users across different domains happens infrequently in real-world business applications. For example, there might be only a limited number of users who purchased books and digital music together on Amazon. Therefore, it is important to develop cross-domain recommendation methods that work for the cases dealing with only few such overlap users across the domains.

We also point out that most of the existing models for cross-domain recommendations only focus on the explicit information of users and items, without considering latent and complex relations between them. Therefore, they could not effectively include user and item features during the cross-domain recommendation process, thus limiting the performance of such systems. Latent embedding methods, on the other hand, constitute a powerful tool to extract latent user preferences from the data record and effectively model users and items (Zhang et al. 2019). Therefore, we focus on modeling users in the latent, rather than the feature space, in our dual-learning recommendation approach.

In this paper, we apply the idea of dual learning to cross-domain recommendations using latent embeddings of user and item features, as shown in Figure 1. We assume that if two users have similar preferences in a certain domain, their preferences should also be similar across other domains. We address this assumption by proposing a

unifying mechanism that extracts the essence of preference information in each domain and improves the recommendations for both domains *simultaneously* through better understanding of user preferences. Specifically, the proposed dual learning model learns latent orthogonal mappings across the two domains and provides cross-domain recommendations leveraging user preferences from both domains. Furthermore, we empirically demonstrate that by iteratively updating the dual recommendation model, we simultaneously improve recommendation performance over both domains and surpass all baseline models.

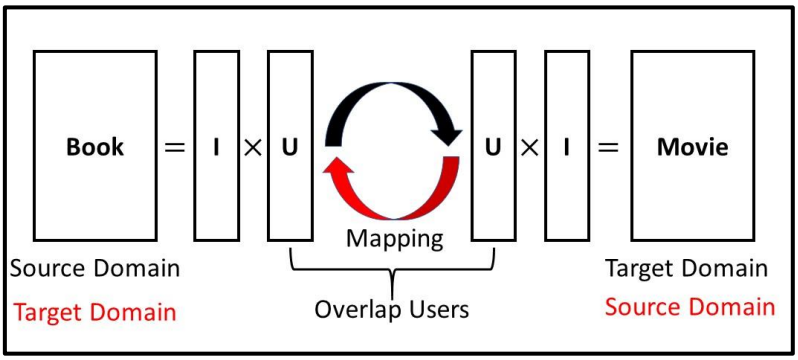

Figure 1. Dual learning framework for cross-domain recommendations

In this paper, we make the following contributions:

(1) We propose to apply the combination of dual learning mechanism and latent embedding approach to the problem of cross domain recommendations.

(2) We empirically demonstrate that the outlined approach outperforms the state-of-the-art methods and improves recommendation accuracy across multiple domains and experimental settings.

(3) We theoretically demonstrate the convergence condition for the simplified case of our model and empirically show that the proposed model stabilizes and converges after several iterations.

(4) We illustrate that the proposed model can significantly reduce the need for overlap information and can also be easily extended to multiple-domain

recommendation applications. Therefore, the proposed model is especially useful for large online platforms, such as Amazon and Alibaba.

Literature Review

Cross Domain Recommendations

Cross domain recommendation approach (Cantador et al. 2015) constitutes a powerful tool to deal with the data sparsity problem. Typical cross domain recommendation models are extended from single-domain recommendation models, including CMF (Singh and Gordon 2008), CDCF (Li et al. 2009; Hu et al. 2013), CDFM (Loni et al. 2014), Canonical Correlation Analysis (Sahebi et al. 2017), Dual Regularization (Wu et al. 2018) and transfer learning approaches (Hu et al. 2018; Lian et al. 2017). These models have achieved great success in addressing the cold-start problem and enhancing recommendation performance.

However, these models do not fundamentally address the relationship between different domains, for they do not improve recommendation performance of both domains simultaneously, thus might not release the full potential of utilizing the cross-domain user interaction information. Also, they do not explicitly model user and item features during recommendation process. Different from the method in (Li and Tuzhilin 2020), which requires massive amount of overlap information, in this paper we propose a novel dual-learning based model combining with autoencoder to overcome these issues and significantly improve recommendation performance.

Dual Learning

Transfer learning (Pan & Yang 2009) deals with the situation where the data obtained from different resources are distributed differently. It assumes the existence of common

knowledge structure that defines the domain relatedness and incorporate this structure in the learning process by discovering a shared latent feature space in which the data distributions across domains are close to each other and subsequently learning the latent factors and associations spanning in that shared subspace.

In addition, to exploit the duality between these two distributions and to enhance the transfer capability, researchers propose the dual transfer learning mechanism (Long et al. 2012) that simultaneously learns the marginal and conditional distributions. Recently, researchers manage to achieve great performance on machine translation with dual-learning mechanism (He et al. 2016; Xia et al. 2018). All these successful applications address the importance of exploiting the duality for mutual reinforcement. However, none of them apply the dual transfer learning mechanism into cross-domain recommendation problems, where the duality lies in the symmetrical correlation between source domain and target domain user preferences. In this paper, we utilize a novel dual-learning mechanism to improve recommendation performance.

Method

In this section, we present the proposed dual learning model for cross-domain recommendations. First, we construct feature embeddings from the explicit user and item features using the autoencoding technique. Then we design a specific type of latent orthogonal mapping function for transferring feature embeddings across two different domains through dual transfer learning mechanism. Moreover, we theoretically demonstrate convergence of the proposed model under certain assumptions, as well as its ability to reduce the overlap information requirement. The complete architecture of the dual learning system is illustrated in Figure 2. We will explain the details in the following sections.

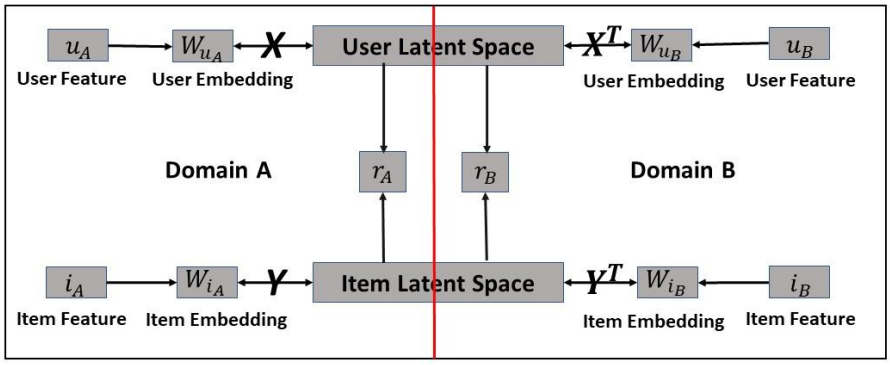

Figure 2. Dual learning framework for cross-domain recommendations

Feature Embeddings

To effectively extract latent user preferences and efficiently model features of users and items, we utilize an autoencoder framework that learns the latent representations of user and item features and transforms the heterogeneous and discrete feature vectors into continuous feature embeddings. It has multiple important advantages over the classical feature-based approaches, as discussed in literatures on deep-learning based recommender systems (Zhang et al. 2019; Wang et al. 2015; He et al. 2017).

We denote the feature information for user u as $u = \{u_1, u_2, \dots, u_m\}$ and the feature information for item i as $i = \{i_1, i_2, \dots, i_n\}$, where m and n stand for the dimensionality of user and item feature vectors respectively. The goal is to train two separate neural networks: the encoder network that maps feature vectors into latent embeddings, and the decoder network that reconstructs feature vectors from latent embeddings. Due to effectiveness and efficiency of the training process (Rumelhart et al. 1985), we formulate both the encoder and the decoder as multi-layer perceptron (MLP). MLP learns the hidden representations by optimization reconstruction loss:

$$L_{AE} = MLP_{decoder}(MLP_{encoder}(u, i))$$

where $MLP_{encoder}$ and $MLP_{decoder}$ represents the MLP network for encoder and decoder respectively. The user embeddings and item embeddings are subsequently

obtained by $W_u = MLP_{encoder}(u)$ and $W_i = MLP_{encoder}(i)$. In this step, we train the autoencoder separately for users and items in different domains to avoid the information leak between domains.

Note that regularization technique is typically crucial to the feasibility and efficient optimization of the metric learning model. Especially when the data points are distributed in a high-dimensional space, they might spread too widely because of the curse of dimensionality and thus become ineffective to be mapped into the same shared latent space. Therefore, we restrict all the user and item embeddings W_u and W_i within the unit ball, i.e., $||W_u||^2 < 1$ and $||W_i||^2 < 1$, to ensure the robustness of the learned metric. Therefore, we do not need to impose additional regularization during the optimization of the subsequent metric learning process.

Latent Orthogonal Mapping

To effectively enable the bidirectional dual learning process between the source domain and the target domain, in this paper we introduce a novel latent orthogonal mapping function. The fundamental assumption of the proposed approach is that, if two users have similar preferences in the source domain, their preferences should also be similar in the target domain. This implies that we need to preserve similarities between users during the dual learning across domains. In particular, we restrict the learning process as the latent orthogonal mapping for two reasons. First, it preserves similarities between user embeddings across different latent spaces, since orthogonal mapping preserves inner product of vectors (Meyer 2000). Second, it automatically derives the inverse mapping function as its own transpose, because $I = XX^T$ holds for any given orthogonal matrix X (Meyer 2000), thus making inverse mapping function equivalent to its transpose. This simplifies the learning procedure and reduces the complexity of

the proposed model. Orthogonality restricts the parameter space of the transitional mapping and thus requires less overlap user information to determine this mapping function, as we will discuss in the next section. Using the latent orthogonal mapping, we could transfer user preferences from one domain to the other without losing or distorting information during the process.

Dual Learning

To improve efficiency and effectiveness of recommender systems, we propose to apply dual learning mechanism to cross-domain recommendation tasks, where we transfer user preferences between the source domain and the target domain simultaneously. Consider two different domains D_A and D_B that contain user-item interactions as well as user and item features. In real business practices, user group in two different domains often overlaps with each other, meaning that some users may have purchased items in both domains. Meanwhile, there is no overlap of item between two domains, as each item only belongs to one single domain, such as books or movies. One crucial assumption we make here is that, if two users have similar preferences in D_A, they are supposed to have similar preferences in D_B as well, indicating that we can learn and improve user preferences in both domains simultaneously over time. Therefore, to obtain better understanding of user preferences in D_A, we could utilize external user preference information from D_B and combine them together. Similarly, we could get better recommendation performance in D_B if we utilize user preference information from D_A at the same step. To leverage duality of the cross-domain recommendation models and to improve effectiveness of both tasks simultaneously, we conduct dual transfer learning recommendations for the two models together and learn the latent mapping function accordingly.

Specifically, we denote the user embeddings for domain D_A and D_B as W_{u_A} and W_{u_B} that we obtained in the previous section. We also denote the overlap users between two domains as $ou_A = ou_B$, and their user embeddings W_{ou_A}, W_{ou_B} correspondingly. The goal of proposed dual learning model is to find the optimal mapping matrix X to minimize the sum of squared Euclidean distances between the mapped user embeddings XW_{ou_A} and the target user embeddings W_{ou_B} for the same overlap users:

$$L_{o_A} = argmin_X \sum_{\{W_{ou_A}, W_{ou_B}\} \in \{ou_A, ou_B\}} \left| XW_{ou_A} - W_{ou_B} \right|^2$$

This optimization is equivalent to its dual form:

$$L_{o_B} = argmin_X \sum_{\{W_{ou_A}, W_{ou_B}\} \in \{ou_A, ou_B\}} \left| W_{ou_A} - X^T W_{ou_B} \right|^2$$

Similarly, we could also identify items closely related to each other across two domain as $si_A = si_B$, for example, the novel and the movie based on that novel. Their item embeddings are denoted as W_{si_A} and W_{si_B} correspondingly. The latent orthogonal mapping Y aims minimizes the Euclidean distances in the latent space:

$$L_{s_A} = argmin_Y \sum_{\{W_{si_A}, W_{si_B}\} \in \{si_A, si_B\}} \left| YW_{si_A} - W_{si_B} \right|^2$$

This optimization is equivalent to its dual form:

$$L_{s_B} = argmin_Y \sum_{\{W_{si_A}, W_{si_B}\} \in \{si_A, si_B\}} \left| W_{si_A} - Y^T W_{si_B} \right|^2$$

We constrain the dual learning mappings X and Y to be orthogonal mappings, which serve to enforce structural invariance of user preferences in each domain, while preventing a degradation in mono-domain recommendation performance for learning better transitional mappings. We optimize all loss functions simultaneously for the dual learning process. The proposed dual learning algorithm for providing cross-domain recommendations is summarized in Algorithm 1.

Algorithm 1 Dual Learning for Cross-Domain Recommendations

Input: User Embeddings W_{u_A} and W_{u_B}; Item Embeddings W_{i_A} and W_{i_B}; Ratings r_A and r_B; Neural Recommender Systems RS_A and RS_B; Orthogonal Mappings X and Y; Overlap Users $ou_A = ou_B$; and Similar Items $si_A = si_B$
repeat
  for (W_{u_A}, W_{i_A}, r_A) in A, (W_{u_B}, W_{i_B}, r_B) in B **do**
    Estimate ratings in A as $r'_A = RS_A(XW_{u_A}, YW_{i_A})$
    Backpropogate $L_A = ||r_A - r'_A||$, Update RS_A
    Estimate ratings in B as $r'_B = RS_B(W_{u_B}, W_{i_B})$
    Backpropogate $L_B = ||r_B - r'_B||$, Update RS_B
  end for
  for W_{ou_A}, W_{ou_B} in Overlap Users **do**
    Backpropogate $L_{o_A} = ||XW_{ou_A} - W_{ou_B}||$, Update X
    Backpropogate $L_{o_B} = ||X^T W_{ou_B} - W_{ou_A}||$, Update X
    Orthogonalize X
  end for
  for W_{si_A}, W_{si_B} in Similar Items **do**
    Backpropogate $L_{s_A} = ||YW_{si_A} - W_{si_B}||$, Update Y
    Backpropogate $L_{s_B} = ||Y^T W_{si_B} - W_{si_A}||$, Update Y
    Orthogonalize Y
  end for
until convergence

Theoretical Analysis

Convergence Analysis

In this section, we present the convergence theorem of the recommendation model, especially matrix factorization method with dual learning mechanism. We denote the rating matrices as R_A and R_B for the two domains A and B respectively. The goal is to find the approximate non-negative factorized matrices to minimize the reconstruction loss of the original rating matrices. To do so, in Algorithm 1 we perform multiplicative update algorithms (Lin 2007) to optimize the reconstruction loss iteratively for the two domains as follows:

$$L_A = ||R_A - W_A H_A|| + ||R_A - X^T W_B Y^T H_B||$$

$$L_B = ||R_B - W_B H_B|| + ||R_B - X W_A Y H_A||$$

Note that $||R_A - W_A H_A||$ and $||R_B - W_B H_B||$ are guaranteed to converge based on the analysis in the existing literature (Lee and Seung 2001) for the classical single-domain matrix factorization. In addition, we recall that X and Y are orthogonal matrices with bounded values. Therefore, following the analysis in (Li and Tuzhilin 2020), we know

that $||R_A - X^T W_B Y^T H_B||$ and $||R_B - X W_A Y H_A||$ are also guaranteed to converge. To sum up, we show that the repeated iteration of update rules is indeed guaranteed to converge to a locally optimal solution for the dual learning method.

In this paper, to capture latent interactions between users and items, we use the neural network-based matrix factorization approach, therefore it still remains unclear if this convergence happens to the proposed model. However, our hypothesis is that even in our case we will experience similar convergence process guaranteed in the classical matrix factorization case. We validate this hypothesis through extensive experiments in the next section.

Reducing the need for overlap information

By framing the cross-domain recommendation problem as the dual learning problem, we alleviate the need to extract explicit user behaviors from both domains; instead, we only need to learn the latent orthogonal mapping that represents the relations between the two domains. In addition, by mapping both user and item features into the latent space, we overcome the problem of discretization in the feature space and mathematically formulate the problem as follows: given a set of k-dimensional feature vectors $\{W_{A_1}, W_{A_2}, \ldots, W_{A_k}\}$ in domain A and $\{W_{B_1}, W_{B_2}, \ldots, W_{B_k}\}$ in domain B, the goal is to identify the matrix X that $XW_{A_i} = W_{B_i}$ holds for every i from 1 to k. In addition, we restrict this matrix X to be orthogonal, i.e., $XX^T = I$. From basic linear algebra (Meyer 2000), we know that the unique solution exists when the number of restricted equations $n = \frac{k(k-1)}{2}$. Therefore, when the dimension of feature embeddings is k, we only need $\frac{k(k-1)}{2}$ number of overlap users to determine the metric learning mapping. For example, when k=16, we only need to identify 120 overlap users between two domains. Experiment results show that the proposed model could achieve

comparable recommendation performance using as little as 8 overlap users.

Experiment

Dataset

We collect a large-scale anonymized dataset from a European online recommendation service over the period of 2011 to 2014 for carrying our experiments. It allows users to rate and review a range of items from various domains, while each domain was treated as an independent sub-site with separate within-domain recommendations. Consequently, the combination of explicit user feedback and diverse domains makes the dataset unique and valuable for cross-domain recommendations. We select the subset that includes three largest domains - books, movies and music, three domains being linked together through a common user ID identifying the same user. We normalize the scale of ratings to 0 and 1. Note that each user might purchase items from different domains, while each item only belongs to one single domain. The basic statistics about this dataset are shown in Table 1.

Domain	Book	Movie	Music
#Users	804,285	959,502	45,962
#Items	182,653	79,866	183,114
#Records	223,007,805	51,269,130	2,536,273
Density	0.0157%	0.0669%	0.0301%

Table 1: Descriptive Statistics of the Datasets

Baseline Models and Evaluation Metrics

We compare our proposed dual learning model with a group of state-of-the-art cross-domain recommendation methods based on the standard metrics of RMSE, MAE, Precision and Recall measures (Herlocker et al. 2004).:

- **CCCFNet** (Lian et al. 2017) Cross-domain Content-boosted Collaborative Filtering neural NETwork (CCCFNet) utilizes factorization to tie CF and

content-based filtering together with a unified multi-view neural network.

- **CDFM** (Loni et al. 2014) Cross Domain Factorization Machine (CDFM) proposes an extension of FMs that incorporates domain information in this pattern, which assumes that user interaction patterns differ sufficiently to make it advantageous to model domains separately.
- **CoNet** (Hu et al. 2018) Collaborative Cross Networks (CoNet) enables knowledge transfer across domains by cross connections between base networks.
- **NCF** (He et al. 2017) Neural Collaborative Filtering (NCF) is a neural network architecture to model latent features of users and items using collaborative filtering method. The NCF models are trained separately for each domain without transferring any information.
- **CMF** (Singh et al. 2008) Collective Matrix Factorization (CMF) simultaneously factor several matrices, sharing parameters among factors when a user participates in multiple domains.

Results

Cross-Domain Recommendation Performance

Since we have three different domains, i.e. books, movies and music, this results in three domain pairs for evaluation. As shown in Table 2 (a-c), the proposed dual learning model significantly and consistently outperforms all other baselines in terms of RMSE, MAE, Precision and Recall metrics across all the three domain pairs. For example in Table 2(a), we show that the RMSE measures for book vs, movie domains obtained using the dual learning model outperform the second-best baselines by 5.77% and 12.60%, while the MAE measures outperform by 7.64% and 15.57%, the Precision

measures outperform by 7.35% and 5.84%, and the Recall measures outperform by 11.23% and 4.16% respectively. We observe similar significant improvements of cross-domain recommendations in book vs. music domains and movie vs. music domains.

To summarize, all these results show that the proposed dual learning model works well in practice, and indeed achieves significant improvements over state-of-the-art cross-domain recommendation baselines. It is also worth noticing that the dual learning model only requires the same level of time complexity for training compared with other baseline models.

Algorithm	Book Domain				Movie Domain			
	RMSE	MAE	Pre@5	Rec@5	RMSE	MAE	Pre@5	Rec@5
Dual Learning	0.2172*** (0.0013)	0.1609*** (0.0009)	0.8971*** (0.0029)	1.0000*** (0.0000)	0.2102*** (0.0013)	0.1600*** (0.0009)	0.9077*** (0.0031)	1.0000*** (0.0000)
CML	0.2408	0.1927	0.8150	0.8864	0.2342	0.1924	0.8520	0.9380
CCCFNet	0.2639	0.1841	0.8102	0.8872	0.2476	0.1939	0.8545	0.9300
CDFM	0.2494	0.2165	0.7978	0.8610	0.2289	0.1901	0.8498	0.9312
CMF	0.2921	0.2478	0.7972	0.8523	0.2738	0.2293	0.8324	0.9012
CoNet	0.2305	0.1892	0.8328	0.8990	0.2298	0.1903	0.8680	0.9601
NCF	0.2315	0.1887	0.8357	0.8924	0.2276	0.1895	0.8428	0.9495

Table 2(a): Cross Domain Recommendation Performance on Book-Movie Domain Pairs

Algorithm	Book Domain				Music Domain			
	RMSE	MAE	Pre@5	Rec@5	RMSE	MAE	Pre@5	Rec@5
Dual Learning	0.2147*** (0.0012)	0.1608*** (0.0009)	0.8725*** (0.0028)	0.9692*** (0.0014)	0.2680*** (0.0013)	0.2242*** (0.0014)	0.8550*** (0.0027)	0.9101*** (0.0011)
CML	0.2398	0.1902	0.8112	0.9120	0.3075	0.2478	0.7922	0.8402
CCCFNet	0.2630	0.1842	0.8150	0.9108	0.3090	0.2422	0.7902	0.8388
CDFM	0.2489	0.2155	0.8104	0.9102	0.3252	0.2463	0.7895	0.8365
CMF	0.2921	0.2478	0.8072	0.8978	0.3478	0.2698	0.7820	0.8324
CoNet	0.2307	0.1897	0.8230	0.9300	0.2801	0.2410	0.7912	0.8428
NCF	0.2315	0.1887	0.8357	0.8924	0.2828	0.2423	0.7930	0.8450

Table 2(b): Cross Domain Recommendation Performance on Book-Music Domain Pairs

Algorithm	Movie Domain				Music Domain			
	RMSE	MAE	Pre@5	Rec@5	RMSE	MAE	Pre@5	Rec@5
Dual Learning	0.2103*** (0.0012)	0.1633*** (0.0009)	0.9078*** (0.0029)	1.0000*** (0.0000)	0.2707*** (0.0013)	0.2291*** (0.0014)	0.8488*** (0.0027)	0.9100*** (0.0011)
CML	0.2259	0.1870	0.8578	0.9270	0.3018	0.2590	0.7875	0.8390
CCCFNet	0.2468	0.1932	0.8398	0.9310	0.3090	0.2433	0.7952	0.8498
CDFM	0.2289	0.1895	0.8306	0.9382	0.3252	0.2467	0.7880	0.8460
CMF	0.2738	0.2293	0.8278	0.9222	0.3478	0.2698	0.7796	0.8400
CoNet	0.2302	0.1908	0.8450	0.9508	0.2811	0.2428	0.8010	0.8512
NCF	0.2276	0.1895	0.8428	0.9495	0.2828	0.2423	0.7930	0.8450

Table 2(c): Cross Domain Recommendation Performance on Movie-Music Domain Pairs

Number of Overlap Users

Besides the superior recommendation performance, we claim that the proposed model also works well with very few overlap users. To validate this claim, we include different number of overlap users across the domain pairs and provide cross-domain recommendations through the proposed dual learning model.

As shown in Figure 3, when we only include 8 overlap users to provide cross-domain recommendations, we could achieve the same level of recommendation performance as the state-of-the-art baseline models, which utilizes all the overlap users. Specifically, in the Book-Movie recommendation scenario, we witness only 0.23\% decrease in RMSE, 0.59\% decrease in MAE, 1.52\% increase in Precision and 0.68\% increase in Recall, where all these changes are not statistically significant. We also observe similar results in the other two experiments. In this way, we empirically show that the proposed model works well with very few overlap users.

In addition, we show in Figure 3 that the more overlap users we include in the model, the better recommendation performance we obtain. When we start to include a few overlap users into the cross-domain recommendation model, we observe rapid and significant improvements of recommendation performance. After we add more overlap users into the model, we witness the phenomenon of "elbow point", where additional overlap users only incrementally improve the performance. Eventually when we utilize all overlap users in the datasets, we achieve state-of-the-art recommendation performance and outperforms all other baselines consistently and significantly.

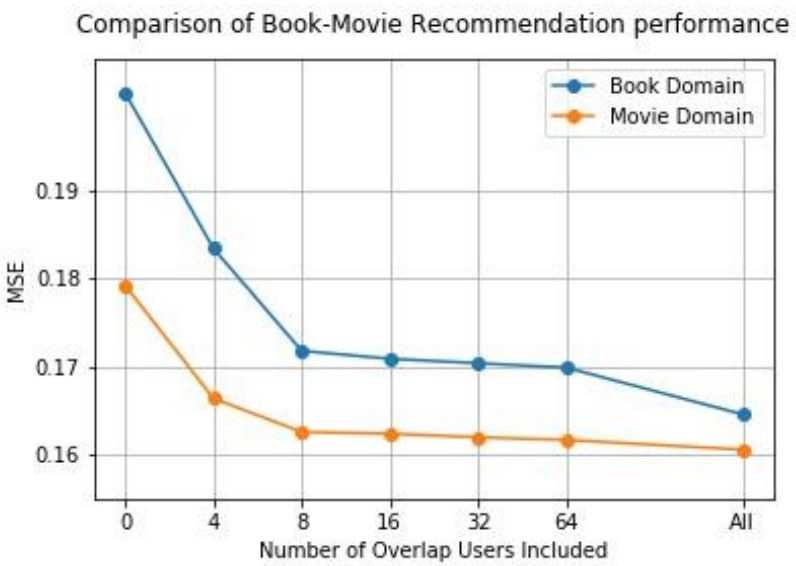

Figure 3: MSE metric of Dual Learning with Different Number of Overlap Users

Convergence

In the theoretical analysis section, we proof convergence of the dual learning method for classical matrix factorization problem. Note that however, this proposition is not applicable to our proposed model because the optimization process is different. Therefore, we test the convergence empirically in our study, and conjecture should still happen even though our method does not satisfy the condition. We conduct the convergence study to empirically demonstrate the convergence.

In particular, we train the model iteratively for 100 epochs until the change of both loss function are less than 1e-5. We plot the training loss over time in Figure 4 for the book domain, where we observe that the proposed dual learning model starts with relatively higher loss due to the noisy initialization, but manages to stabilize quickly and significantly outperforms the NCF baseline only after 10 epochs. We could observe similar phenomenon in the epoch-loss plots for the movie and music domains as well. Therefore, we empirically validate that the proposed dual learning model indeed converges in practice.

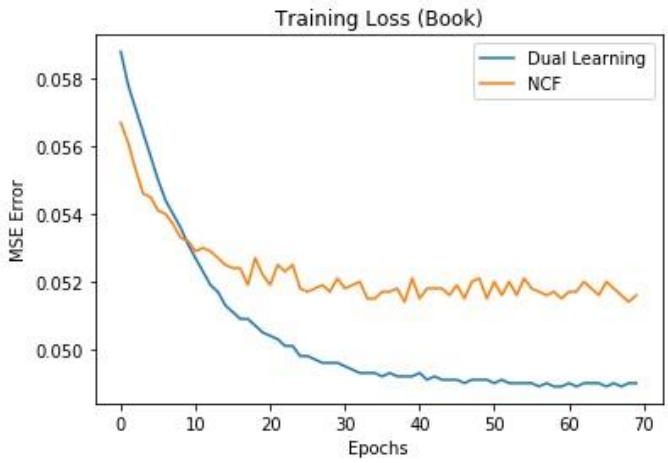

Figure 4: Epoch-Loss Plot for the Book Domain

Discussions and Conclusions

In this paper, we present a dual learning based cross-domain recommendation model that significantly improves recommendation performance across different domains and reduces the need for the overlap users. We accomplish this by transferring latent information from one domain to the other through embeddings and iteratively going through the transfer learning loop until the models stabilize for both domains. We also prove that this convergence is guaranteed under certain conditions and empirically validate the hypothesis for our model across different experimental settings.

Note that, the presented approach provides several benefits, including that it (a) transfers information about latent interactions instead of explicit features from the source domain to the target domain; (b) utilizes the dual learning mechanism to enable the bidirectional training that improves performance measures for both domains simultaneously and reduces the need for overlap information; (c) learns the latent orthogonal mapping function across the two domains, that (i) preserves similarity of user preferences and thus enables proper transfer learning and (ii) computes the inverse mapping function efficiently. Therefore, the proposed model is especially useful for large online platforms, such as Amazon and Alibaba.

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